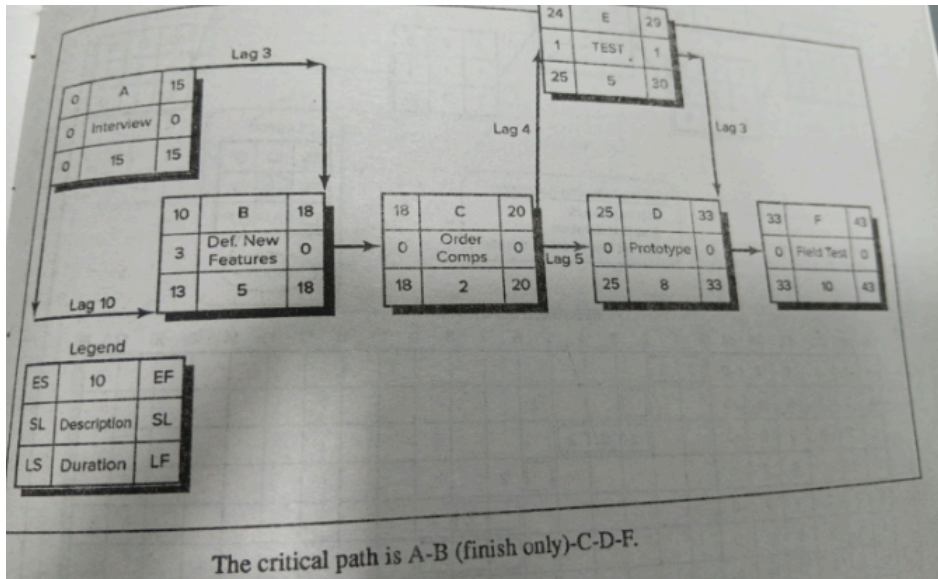


Confirmation for Free Float Formulas In FSPM

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FORMULAS

Total Float (TF)

Maximum time an activity can be delayed without delaying the project finish

$$\text{Total Float (TF)} = \text{LS} - \text{ES} = \text{LF} - \text{EF}$$

Free Float (FF)

Time an activity can be delayed without delaying the Early Start of the next activity

$$\text{Free Float (FF)} = \text{ES}_{\text{successor}} - \text{EF}_{\text{current}} - \text{Lag}$$

If multiple successors exist → take **minimum** value.

ACTIVITY-WISE CALCULATION (from diagram)

Activity A (Interview)

- ES = 0
- EF = 15
- LS = 0

- $LF = 15$

Total Float

$$TF = LS - ES = 0 - 0 = 0$$

Free Float

- Successor: **B**
- $ES(B) = 10$
- $Lag = 3$

$$FF = 10 - 15 - 3 = -8 \Rightarrow 0$$

→ **TF = 0, FF = 0 (Critical)**

Activity B (Define New Features)

- $ES = 10$
- $EF = 18$
- $LS = 13$
- $LF = 18$

Total Float

$$TF = 13 - 10 = 3$$

Free Float

- Successor: **C**
- $ES(C) = 18$
- $Lag = 0$

$$FF = 18 - 18 - 0 = 0$$

→ **TF = 3, FF = 0**

Activity C (Order Components)

- $ES = 18$
- $EF = 20$
- $LS = 18$
- $LF = 20$

Total Float

$$TF = 18 - 18 = 0$$

Free Float

- Successor: **D**
- $ES(D) = 25$
- $Lag = 5$

$$FF=25-20-5=0$$

→ **TF = 0, FF = 0 (Critical)**

Activity D (Prototype)

- ES = 25
- EF = 33
- LS = 25
- LF = 33

Total Float

$$TF=25-25=0$$

Free Float

- Successor: **F**
- ES(F) = 33
- Lag = 0

$$FF=33-33-0=0$$

→ **TF = 0, FF = 0 (Critical)**

Activity E (Test)

- ES = 24
- EF = 29
- LS = 25
- LF = 30

Total Float

$$TF=25-24=1$$

Free Float

- Successor: **D**
- ES(D) = 25
- Lag = 4

$$FF=25-29-4=-8\Rightarrow 0$$

→ **TF = 1, FF = 0**

Activity F (Field Test)

- ES = 33
- EF = 43
- LS = 33

- $LF = 43$

Total Float

$$TF = 33 - 33 = 0$$

Free Float

- No successor $\rightarrow FF = 0$

$\rightarrow TF = 0, FF = 0$ (Critical)

SUMMARY TABLE

Activity Total Float Free Float

A	0	0
B	3	0
C	0	0
D	0	0
E	1	0
F	0	0

CRITICAL PATH (confirmed)

Activities with $TF = 0$:

$A \rightarrow B \rightarrow C \rightarrow D \rightarrow F$



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